

Analytics Engineer Take-Home Assessment

Alltown Fresh Mobile Ordering, Customer Loyalty
Metrics, and Promotion Measurement

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Thesis

The strongest answer is not one big query or one p-value. It is a governed Data Product with explicit grain, tested transformations, and cautious measurement.

3

Workstreams

1. SQL Refactor
2. Dimensional Modeling
3. Experiment Measurement

2.9x

faster shape

Local Benchmark directionally
favored the Refactor

\$1.01

basket lift

Directional, not significant at $\alpha = 0.05$

Ship the Modeling Refactor, stage the messy data deliberately, and treat the Promotion Result as promising but unproven.

Q1: SQL / dbt

The prototype breaks Customer Grain through Option Fanout and Many-to-Many joins.

Recommendation: Refactor into Staging, Intermediate, and Mart Models with tests at every grain.

Q2: SSOT Modeling

The data is useful but not reporting-ready.

The Staging Layer must handle Source Scope, Identity, Category Cleanup, Orphan Options, and Outlier Flags.

Q3: Measurement

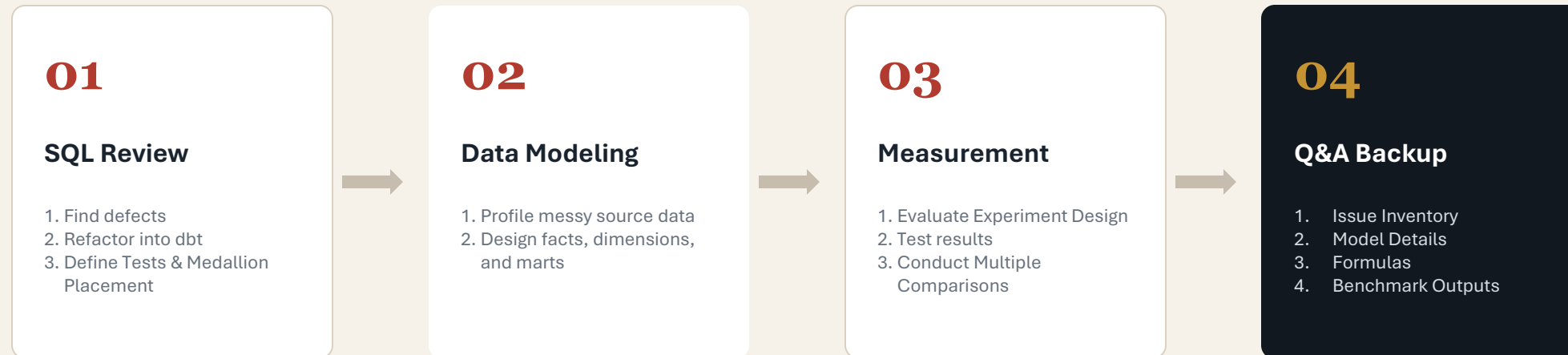
The Promotion produced a +\$1.01 directional Basket Lift, but had a one-tailed p-value of 0.072.

Recommendation: Be cautious until Profit & Validity Checks complete.

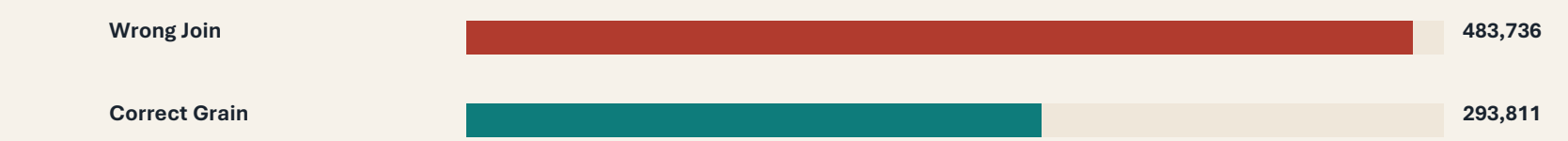
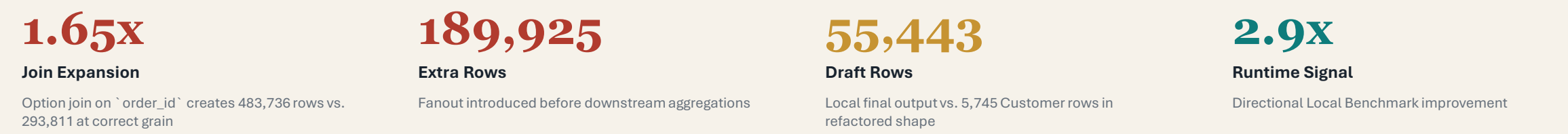
Evaluation Lens

Protect grain. Quantify risk. Separate signal from decision. Keep the business question visible.

The Presentation follows the same order as the Assessment, with Appendix depth reserved for Q&A.



The Prototype produces a Customer Mart that is neither Customer-grain nor Revenue-safe.



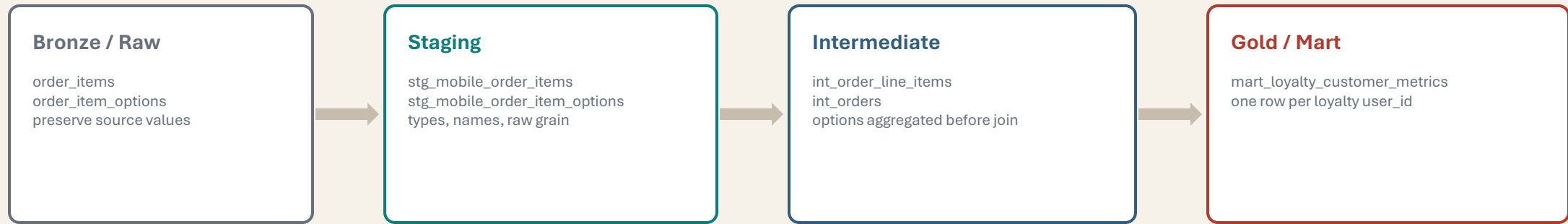
Core Diagnosis: The query joins options at the wrong key, then mixes option, line, order, monthly, and customer-location-card grains before selecting a Customer-looking final table.

The Defects matter because each one changes the Business Meaning of the Output.

Issue	What Is Wrong	Downstream Impact
Option Fanout	Join uses `order_id` only	Inflates rows, category counts, revenue, compute
`NULL` Line Totals	No COALESCE for missing options	Drops unmodified items from SUM(`line_total`)
Spend Logic	SUM(`item_price`) ignores quantity/options	Understates Customer Lifetime Value
Final Grain	Joins multi-row CTEs on `user_id`	BI Users double-count Customers & Spend
Recency	CURRENT_DATE() baked in	Historical rebuilds <u>not</u> reproducible

The Production Fix is less about clever SQL and more about enforced contracts: every model needs a stated grain & tests that fail when that grain breaks.

Split the Monolith into tested layers so every grain is explicit.



Monthly trends become a separate Customer-Month Mart. They do not belong in a one-row-per-Customer table.

The Refactor is only trustworthy if Tests protect the Assumptions the Business will rely on.

Grain

1. Unique `order\_id`
2. Unique `lineitem\_id`
3. Unique customer `user\_id`

Relationships

Options must map to `order\_id` and `lineitem\_id`

Accepted Values

1. Currency
2. Segment Labels
3. Recency Labels
4. Source App

Reconciliation

Customer Lifetime Spend reconciles back to Order Totals

Data Quality

1. Orphan Options
- NULL Loyalty Users
Extreme Order Values

Generic tests catch Schema Breakage. Singular tests catch Business Contract Breakage.

The data is rich enough for Reporting, but the Staging Layer has real work to do first.

203,519

Line-Item Rows

1 row per Mobile Order line-item

193,017

Option Rows

Modifiers selected on line-items

129,511

Alltown Fresh Orders

in-scope Production App Orders

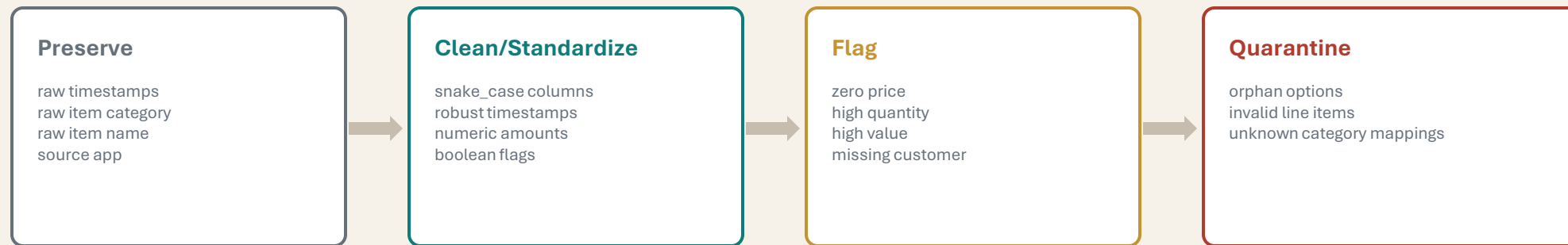
20,174

Distinct Users

Mobile Account Identity, not always Loyalty

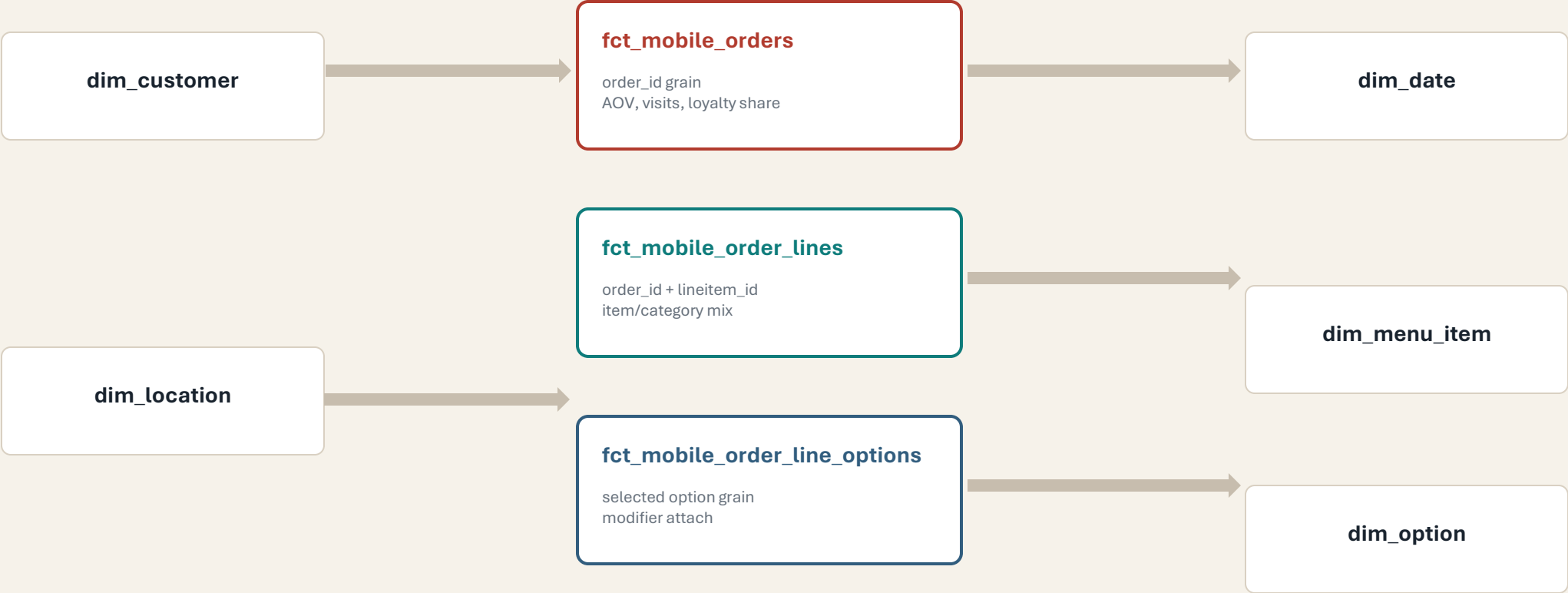
Messiness	Evidence	Handling
Source Scope	1,270 Neighborhood Perks; 826 development rows	Filter in intermediate, preserve raw
Identity	17,808 NULL `user_id`; 28 Loyalty rows NULL	Exclude from Customer Mart, flag exceptions
Categories	URLs, typos, Kids vs Kid's	Canonical Mapping Table
Outliers	\$2.5M max Order Total	Flag, Review, do <u>not</u> silently drop

Treat messy values as modeled facts: Preserve, Clean, Flag, and Quarantine.



This is the practical SSOT move: Analysts get clean marts, Engineers keep traceability, and Business Users can see what was excluded or flagged.

A Star Schema protects Transaction Grain while giving Tableau clean marts.



The Model enables Stakeholder Questions without forcing Analysts back into raw CSV logic.

Question	Models	Metrics
How is mobile ordering performing by location/time?	orders, location daily mart	orders, sales, AOV, loyalty share, daypart mix
Which categories and items drive sales?	order lines, item daily mart	units, item sales, category mix, share of sales
How do loyalty customers differ?	orders, customer mart	repeat rate, lifetime spend, recency, AOV
Which modifiers increase basket value?	options fact, line fact	attach rate, option revenue, basket with/without modifiers
What anomalies require investigation?	staging exceptions, facts	orphan options, zero-price items, outlier orders

The Site-level Test is reasonable, but validity depends on Balance, Execution, and Contamination Control.

Balance

14 stores per group may not balance pre-period Basket, Visits, Coffee Attach, Loyalty Count, or Product Mix.

Parallel Trends

Treatment stores must not already be growing faster before launch.

Spillover

Customers can cross-shop nearby stores & learn about the Promotion.

Execution

1. App/PoS Configurations
2. Signage
3. Employee Training
4. Control-store Suppression

Power

With Site-level randomization, effective n is 28 stores, not all transactions.

Measurement

Define Visits, Active Loyalty Members, Basket Size, Discounts, and Coffee Attach *before* launch.

Measure Non-Coffee Lift & Payback separately from the direct Coffee Giveaway.

Isolate Cross-Sell

1. Exclude coffee items, free redemptions, and coffee discounts from the cross-sell outcome.
2. Compare Non-Coffee Spend per Loyalty Member & Non-Coffee Attach Rate on coffee orders.

Evaluate Payback

Incremental Profit = Incremental Gross Profit from Purchases - Free Coffee Reward Cost - Operating & Marketing Costs

Revenue Lift answers whether Baskets moved. Profit Lift answers whether the Promotion deserves Rollout.

TEST RESULTS

Average Basket Size moved in the right direction, but the evidence does not clear the 95% bar.

+\$1.266

Treatment mean change

test basket - pre basket

+\$0.261

Control mean change

test basket - pre basket

+\$1.005

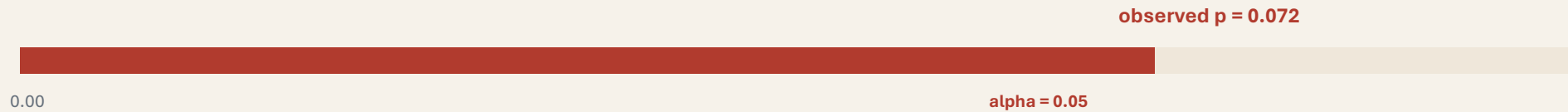
Estimated lift

difference in changes

p = 0.072

one-tailed test

not significant at alpha = 0.05



Conclusion: Directionally positive, but not enough to make a confident go decision alone.

Before Rollout, I would pressure-test Power, Balance, Outliers, Weighting, Mechanism, and Profit.

Power

1. Only 28 site-level observations ($n = 28$)
2. Modest effects hard to detect

Balance

Treatment pre-period Basket was lower than Control;
validate comparability

Sensitivity

Check high-influence stores & Loyalty-Member-weighted results

Mechanism

Decompose Visits, Paid Items, Free Coffee, and Non-Coffee Attach

Economics

Convert Lift → Gross Profit after Reward & Operating Costs

Execution

1. Confirm Treatment Activation
2. **No** Control Contamination

Two "Significant Findings" out of twelve are exploratory until adjusted for False Positives.

$$P(\geq 1 \text{ False Positive}) = 1 - (1 - \alpha)^m$$

With alpha = 0.05 and m = 12:

46%

family false-positive risk

Recommended Framework

1. Pre-specify one Primary Decision Metric, a few secondary diagnostics, and treat the rest as exploratory.
2. Use Bonferroni for conservative family-wise control or Benjamini-Hochberg for False Discovery Rate Control.

Bonferroni Threshold: $\alpha_{adjusted} = \frac{\alpha}{m} = \frac{0.05}{12} \approx 0.042$

Pre-register the three Metrics that map to Behavior, Cross-Sell, and Economics.

1. Weekly Visits per Loyalty Member

1. Primary Behavioral Metric
2. Tests whether the Free-Coffee Ladder changes Visit Frequency among Loyalty Members.

2. Non-Coffee Spend per Loyalty Member

1. Cross-Sell Metric
2. Excludes coffee so result is not mechanically driven by promoted Item

3. Incremental Gross Profit per Loyalty Member

1. Decision Metric
2. Converts Sales Lift → Margin after Reward Cost & Operating Costs

Everything else can still be analyzed but should be labeled as secondary/exploratory.

Move forward with the Data Product Design; be cautious on Promotion Rollout until Economics & Validity Checks clear.

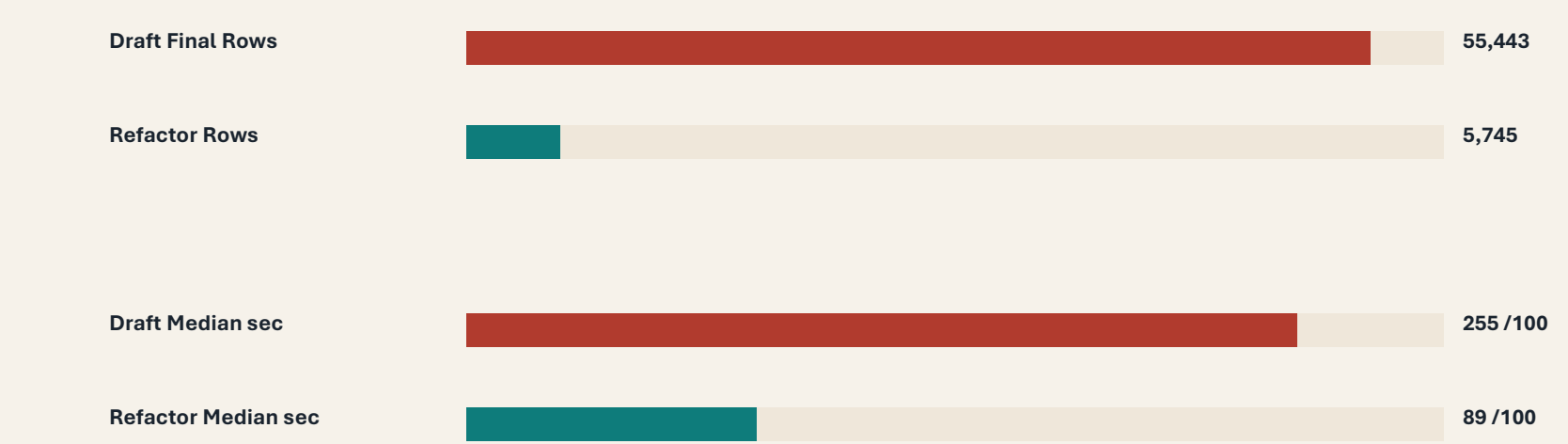
Decision Area	Recommendation	Why
SQL Model	Refactor before Production	Current query breaks grain & revenue logic
SSOT Design	Build facts/dimensions first	Reporting needs tested order, line, option, customer, and location grains
Promotion	Do not call it proven yet	+\$1.01 directional lift, but p = 0.072 one-tailed
Next Iteration	Extend or rerun with pre-registered metrics	Power, economics, and Multiple Comparisons need tighter control

Trustworthy Analytics start with modeled grain & end with decision-ready uncertainty.

Detailed SQL Review Inventory

Risk	Representative Issue	Impact
Fanout	Option join on `order_id` only	Inflated Category Spend and Compute
NULL Math	Missing option makes `line_total` NULL	Unmodified Items excluded from SUMs
Spend	`item_price` not extended by quantity	Lifetime Spend understated
Grain	Customer-Card-Location grouped as Customer	Duplicate Customer rows
Unused Join	Monthly trends joined but not selected	Row Multiplication with no value
Reproducibility	CURRENT_DATE() in recency	Segments drift with Run Date

Benchmark and Row-Count Evidence



Benchmark is directional: local SQLite with indexes, not a Snowflake Warehouse Profile.
In Snowflake, I would validate rows produced per join, bytes scanned, spill, partitions scanned, and credits.

dbt model and Test Inventory

Layer	Models	Tests
Staging	stg_mobile_order_items; stg_mobile_order_item_options	not_null, accepted values, typed fields
Intermediate	int_alltown_fresh_order_line_items; int_alltown_fresh_orders	no fanout, unique grain, relationships
Mart	mart_loyalty_customer_metrics	unique user_id, segment values, spend reconciliation
Custom	singular tests	orphan options, one row per user, lifetime spend reconciliation

Data Quality Profile Details

Signal	Measured Finding	Treatment
App scope	Alltown Fresh plus Neighborhood Perks and development rows	Scope in Intermediate Models
Identity	17,808 NULL `user_id` rows	Retain for Order facts; exclude from Customer marts
Categories	40 raw values with URL artifacts & typos	Mapping Table + Raw-Value Lineage
Options	28 Orphan Option rows	Quarantine Exception Model
Outliers	\$2.5M max order total	Flag & review before KPI Reporting

Mart Model Inventory

Mart	Grain	Primary use
mart_loyalty_customer_metrics	`user_id`	1. Lifetime Value 2. Recency 3. Frequency 4. Top Category
mart_location_daily_mobile_ordering	`restaurant_id` + `date`	Store Performance & Operations Trends
mart_menu_item_performance_daily	`menu item` + `date`	Product/Category Performance
mart_customer_monthly_metrics	`user_id` + `month`	Retention & Cohort Trends

Hypotheses & Statistical Setup

$$H_0 : \mu_{Treatment\ Change} - \mu_{Control\ Change} \leq 0$$

The Promotion did not increase Average Basket Size relative to Control

$$H_A : \mu_{Treatment\ Change} - \mu_{Control\ Change} > 0$$

The Promotion increased Average Basket Size relative to Control.

Excel Test Used

t-Test: Two-Sample Assuming Unequal Variances on each store's change in Average Basket Size

Test Results Tab Output

Statistic	Treatment	Control
Mean Change	1.266429	0.261429
Variance	4.175948	2.032921
Observations	14	14
t Stat	1.50912	
P(T ≤ t) one-tail	0.072442	
P(T ≤ t) two-tail	0.144884	

Interpretation: Positive Directional Lift, not statistically significant at $\alpha = 0.05$.

Multiple Comparisons Math

With 12 tests at $\alpha = 0.05$:

$$P(\text{False Positive} \geq 1) = 1 - (1 - 0.05)^{12} \approx 0.46$$

Framework	Threshold / Interpretation
Unadjusted	$p < 0.05$; Useful for Exploratory Signals but False Positives accumulate
Bonferroni	$p < 0.0042$; Conservative family-wise Error Control
Benjamini-Hochberg	Controls False Discovery Rate; better for ranked Exploratory Metrics
Pre-Registration	One Primary Metric, a few Secondary Metrics, rest exploratory

Assumptions and Q&A Backup

Assumptions

1. Alltown Fresh Production App is the Reporting Scope
2. Snowflake + dbt Core are Target Tools
3. `dim\_date` is available
4. Source data lacks governed Product/Location masters

Q&A Stance

If asked for more time:

1. Run the dbt models in Snowflake
2. Inspect Query Profiles
3. Add Category Reference data
4. Evaluate Gross Margin
5. Rerun Promotion Analysis with richer Customer/Order-level data